

Understanding WiFi Security

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Wi-Fi stands for “wireless fidelity”, a throw back from the audio terminology of “high fidelity”, though the phrase just signifies any network connection that is a wireless transmission based connection.

In simpler terms, think of it as a radio broadcast — except that instead of audio, Wi-Fi transmits data. The typical home user is likely to encounter it when she gets a broadband connection, which terminates into a router at home (like cable TV terminates into a set-top box), only in this case, the cable terminates into a Wi-Fi router.

A Wi-Fi router is like a mini broadcast station that broadcasts in all directions. This means anyone with a receiver (typically a laptop) can receive the broadband connection

Recent events have raised concerns around the usage of wifi. If you know the fundamentals of the technology, it can be a safe and useful tool.

signal, use it, or also eavesdrop on the communication. Think of it like someone splicing a parallel instrument on your telephone line. Now, the rouge instrument can receive or make calls from your line — and for all practical (and law enforcement) purposes it will seem to originate from your phone.

Be aware that all 'hotspots' — public Wi-Fi enabled spots like airports, hotels, etc., are insecure. You may use them for browsing, but do remember that using hotspots for financial transactions can prove disastrous.

So there you go. If you follow the above steps — you should be alright. However, if you are looking for a more robust protection or use your home Wi-Fi for official work, then you need to call in the professionals. Because as they will tell you — these steps may stop a causal attacker — but for a professional, they are only deterrents.

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How To Secure A Home Wi-Fi Connection

So does that make Wi-Fi connections insecure? Well, the short answer is NO, just like a car that is competently driven, is safe! So how does one secure a Wi-Fi connection? Read on.

To begin with, the problem lies with the plug-and-play nature of a typical Wi-Fi router. Most home usage routers are bought over the counter and the manufacturers configure it for novice usage. Which means security configuration is non-existent. Just like say your mobile phone's default pin no. And most users don't bother changing the default configuration. Again like the mobile pin no!

There are few simple steps that one can take to make their Wi-Fi connection safer. Firstly read up the manual. If that is cumbersome (it usually is) — download a two page instruction set on how to secure the router from www.mahindrassg.com or watch a video which shows you how to do it step by step.

Secondly — change the default password for the administrator user of the router. Do this now! You must because — if that is compromised, then all advice that follows is pretty much useless. And don't put easily guessable passwords. So 'Linksys' or 'NetGear' is not a smart choice (these are names of the common routers and a surprising number of users choose them as the administrator password! And while you are at it, enable the firewall on the device.

Thirdly — place your router closer to the center of your home/office to prevent spillage of the signals outside of your physical premises. Some of it will still happen. That is unavoidable. Also switch off the router if you are not using it. Apart from security issues — someone could simply be 'stealing' your connection and you will have to end up paying for that data download.

Ok, now it gets a little technical. Get to the "Wireless Security Setting" page of your router administrative console (or get a teenage nerd to do it for you — make sure you can trust the nerd! — or watch the video mentioned above). Once you are in this console you have to do the following.

Change default SSID or disable it altogether if not required. Turn on WPA encryption and use a strong password there as well. This step will encrypt all traffic passing in your wireless network.

Every network card has a unique address. To find that simply type "ipconfig / all" in your command prompt window. You will see a line called Physical Address followed by a series of 12 digits and alphabets separated by hyphens. Something looking like this: 00-RR-HH-67-54-HF. That is your laptop's physical address. Enter those addresses into the Wi-Fi router and configure it to accept connections only from those addresses.